

SMTP, POP and IMAP — Detailed Lecture Notes

JKSSB Computer Awareness

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38.1 Why email needs protocols

Sending and reading mail involve different programs and different servers. The sender writes a message in a mail program, that program hands it to a mail server, the message travels between servers, waits in the recipient's mailbox, and is finally collected by the recipient's program. For all these independent programs to work together, they follow fixed rules called **protocols**. One protocol sends mail out, and two other protocols bring it in, each in its own way. Secure transmission and protection against mail threats sit on top of these three protocols.

Definition (Email protocol). *A fixed set of rules that a mail program and a mail server follow when sending, forwarding, or collecting mail, so any standard program can work with any standard server.*

Term	Meaning in this lesson
SMTP	Simple Mail Transfer Protocol; the sending protocol.
POP3	Post Office Protocol version 3; a receiving protocol that downloads mail to one machine.
IMAP	Internet Message Access Protocol; a receiving protocol that syncs mail across devices.
SSL/TLS	Secure Sockets Layer / Transport Layer Security; the secure channel for mail transmission.
Spam	Bulk unwanted mail, usually junk or advertising.
Phishing	A fake message impersonating a trusted sender to steal secrets.
Spoofing	Forging the sender address so mail looks as if it came from someone trusted.

The rest of this lesson uses these terms in one consistent sense. **SMTP** always means the sending step. **POP3** always means downloading to one local machine. **IMAP** always means keeping every device in step with the server copy. **SSL/TLS** always means the secure transmission channel. **Spam**, **phishing**, and **spoofing** are three distinct threats defined above.

38.2 SMTP: the sending protocol

SMTP (Simple Mail Transfer Protocol) is the protocol used to send mail. When the sender presses send, the mail program hands the message to the sender's mail server using SMTP, and the servers forward it toward the recipient's mail server using SMTP. SMTP moves the message out of the sender and along the path between servers until it reaches the recipient's mailbox.

Definition (SMTP). *The protocol that sends mail from the sender's program to the mail server and forwards it between mail servers toward the recipient's mailbox.*

Two limits matter for the exam. First, SMTP only *sends*; it does not deliver mail into the recipient's hands for reading. Second, the receiving step that follows is the job of POP3 or IMAP, described next. Any option that says SMTP downloads mail to the inbox, or that POP3 sends mail, reverses the direction and is wrong.

38.3 POP3: download to one machine

POP3 (Post Office Protocol version 3) is a receiving protocol. It pulls mail from the mail server down to one local machine. After the download, the message is typically removed from the server, so the only copy lives on that one computer. Mail downloaded this way can be read offline, but a second device will not see the same mailbox, because the server copy is gone.

Definition (POP3). *A receiving protocol that downloads mail from the server to one local machine, usually removing it from the server.*

POP3 suits a single computer where mail is collected and read in one place. Its weakness is exactly its method: because the mail leaves the server, bookmarks such as read marks, folders, and deletions do not travel to other devices.

Example 38.3.1. *A clerk collects office mail only on the office desktop. With POP3, each morning's mail downloads to that desktop and is removed from the server. The clerk can read it offline at the desk, but the same messages do not appear on the clerk's phone, because the server copy is gone.*

38.4 IMAP: sync across devices

IMAP (Internet Message Access Protocol) is also a receiving protocol, but it works differently from POP3. The mail stays on the server, and the mail program shows a synced view of it: folders, read and unread marks, deletions, and moves are all mirrored between the device and the server. When the reader acts on one device, every other device shows the same change, because each of them reflects the same server copy.

Definition (IMAP). *A receiving protocol that keeps mail on the server and syncs each device's view — folders, read marks, deletions — with that server copy.*

IMAP suits a reader with several devices, such as a phone and a desktop, who must see the same inbox everywhere. Its cost is that the mailbox lives on the server, so reading new mail needs a connection, and the server mailbox must have room.

38.5 POP3 vs IMAP and the full mail flow

Both POP3 and IMAP receive mail; neither sends it. The difference is the method. POP3 downloads to one machine and usually clears the server; IMAP keeps the server copy and syncs every device with it.

Point	POP3	IMAP
Job	Receives mail	Receives mail
Method	Downloads to one machine	Syncs a view of the server copy
Server copy	Usually removed	Kept on the server
Many devices	Poor fit	Designed for it
Offline reading	Natural (copy is local)	Needs a connection for new mail

The full journey of one message combines the sending protocol with one of the receiving protocols. The sender's program sends the message by SMTP to the sender's mail server; that server forwards it by SMTP to the recipient's mail server, where it waits in the mailbox; the recipient's program then collects it with POP3 (download) or IMAP (sync). In one line: sender, then SMTP, then the recipient server, then POP3 or IMAP, then the reader. Sending is always SMTP; the receiving side is POP3 or IMAP.

38.6 To, Cc, BCC, filters, and signatures

The **To** field names the main recipients of a message, and **Cc** (carbon copy) names others kept in the loop. Everyone who receives the message can see both lists. **BCC** (blind carbon copy) names recipients who are hidden from each other: nobody on the To or Cc list sees who is on BCC, and BCC recipients do not see each other. The sender always knows the full list; BCC hides recipients from each other, not from the sender.

Filters, also called **rules**, sort incoming mail automatically. A filter examines the headers and content of each message, such as the sender, the subject, or words in the body, and then files, flags, or deletes matching mail — for example, moving all newsletters to one folder. A mail **signature** is different: it is a fixed block of sender details, such as name, post, and phone number, added at the end of outgoing messages. It identifies the sender but does not encrypt or otherwise protect the message.

38.7 Digital signature vs encryption, and secure transmission

A **digital signature** proves who sent a message and that it was not changed on the way: it gives authenticity and integrity. **Encryption** does something different: it scrambles the message so that only the intended recipient can read it, giving confidentiality. A signed message can still be read by anyone — a signature is not secrecy. An encrypted message can still be forwarded — encryption hides the content but does not by itself prove who sent it. Confusing the two is a tested trap (TRAP-15): a signature never encrypts the body.

Definition (Digital signature vs encryption). *A digital signature provides authenticity (who sent it) and integrity (it was not altered); encryption provides confidentiality (only the intended recipient can read it).*

When mail travels between programs and servers, the channel itself can be protected by **SSL** (Secure Sockets Layer) and its successor **TLS** (Transport Layer Security). Together **SSL/TLS** provide the secure channel over which mail is transmitted, so that others on the path cannot read it. For the exam, the answer is direct: the secure email transmission protocol is SSL/TLS (PYQ-36). The official Junior Assistant item behind PYQ-16 separately confirms the POP3, BCC, filter, and signature-versus-encryption facts used in this lesson.

Example 38.7.1. *An item states, “A digitally signed email can be read only by its recipient.” This is wrong. The signature proves the sender and detects tampering, but anyone can still read the body. Only encryption makes the body readable solely by the intended recipient. The wrong option has assigned encryption’s job to the signature.*

38.8 Spam, phishing, spoofing, and malicious content

Spam is bulk unwanted mail, usually advertising or junk sent to many addresses at once. Its defining mark is that it is bulk and unwanted; it clutters the inbox and may also carry fraud or malware. Spam filters move suspected junk to a separate folder using the sender’s reputation and content rules.

Phishing is a fake message that pretends to be a trusted sender — a bank, an office, or a colleague — in order to steal passwords or money. It typically creates urgency (“verify now”, “account blocked”, “claim within 24 hours”) and asks the reader to click a link or open an attachment and hand over secrets. The defence is to check the real sender address and never enter passwords through a link reached from mail.

Spoofing forges the sender address so that a message looks as if it came from someone trusted. Phishing messages often use spoofing, but the two terms are distinct: phishing is the fraud attempt, and spoofing is the forged-sender trick it may use. A **malicious link** leads to a fake login page or a malware download; hovering over it shows an address that does not match the claimed sender. A **malicious attachment** carries malware behind an innocent-looking file name. The safe rule is to verify an unexpected message through another channel before clicking any link or opening any attachment.

38.9 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. SMTP **sends**; POP3 and IMAP **receive**. Reversing the direction is a common wrong answer.
2. POP3 **downloads** mail to one local machine (usually clearing the server); IMAP **syncs** every device with the server copy. Do not swap them (TRAP-17).
3. A digital signature proves the sender and detects tampering; it does **not** encrypt the body. Encryption alone provides secrecy (TRAP-15).
4. BCC hides recipients **from each other**; the sender always knows the full list, and To/Cc are visible to all (PYQ-16).
5. Secure transmission of email is **SSL/TLS** (PYQ-36). A signature is not a transmission protocol.
6. Spam is **bulk unwanted** mail; phishing is **fake trusted mail that steals secrets**; spoofing is the **forged sender address** either may use.

Example 38.9.1. *An item asks which protocol suits reading the same office mailbox from a phone and a desktop. Trace the methods: POP3 downloads to one machine and clears the server, so the second device misses the mail; IMAP keeps the server copy and syncs every device with it. The answer is IMAP. If the stem instead asks which protocol sends the message out, the answer is SMTP, and if it asks which channel makes transmission secure, the answer is SSL/TLS.*

Exercises

1. State the full form of SMTP, POP3, IMAP, SSL, and TLS, and give the job of each of the first three.
2. Explain the difference between POP3 and IMAP when the same mailbox is read from two devices.
3. Trace one message from the sender to the reader, naming the protocol used at each step.
4. Distinguish a digital signature from encryption, and state which protocol secures email transmission.
5. Define spam, phishing, and spoofing, and state the safe rule for unexpected links and attachments.

Solutions

1. SMTP is Simple Mail Transfer Protocol and sends mail; POP3 is Post Office Protocol version 3 and downloads received mail to one machine; IMAP is Internet Message Access Protocol and syncs received mail across devices; SSL is Secure Sockets Layer and TLS is Transport Layer Security, together the secure transmission channel.
2. With POP3 the mail downloads to the first machine and usually leaves the server, so the second device does not see it; with IMAP the mail stays on the server and both devices show the same synced view of folders, read marks, and deletions.
3. The sender's program sends the message by SMTP to the sender's server; the servers forward it by SMTP to the recipient's server, where it waits; the recipient collects it with POP3 (download) or IMAP (sync).
4. A digital signature provides authenticity and integrity (who sent it, unaltered); encryption provides confidentiality (only the recipient can read it). Secure transmission uses SSL/TLS.
5. Spam is bulk unwanted mail; phishing is a fake trusted message that steals secrets through urgency plus a link or attachment; spoofing forges the sender address. Verify an unexpected message by another channel before clicking any link or opening any attachment.